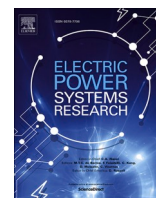




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journal homepage: www.elsevier.com/locate/epsrNew data-driven approach to bridging power system protection gaps with deep learning[☆]Rui Fan^a, Tianzhixi Yin^b, Kun Yang^a, Jianming Lian^{*,c}, John Buckheit^b^a University of Denver, Denver, CO 80208, USA^b Pacific Northwest National Laboratory, Richland, WA 99352, USA^c Oak Ridge National Laboratory, Oak Ridge, TN 37830, USA

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ABSTRACT

Protection is a critical function in power systems to avoid equipment damage, maintain personnel safety, and support system reliability. The existing protection technology that mainly relies on protective relays is not capable of providing 100% protection for all system events. Whenever protective relays fail to respond or respond incorrectly, it leads to the protection gap. In this paper, a new data-driven approach is proposed to complement the traditional protection technology so that the faulty conditions can be adequately distinguished from the transients resulted from normal operations. The proposed approach combines convolutional neural network and long short-term memory (CNN-LSTM) to develop a deep neural network that achieves the invariance in data translation and captures the temporal correlation of input data in time series. It can accurately detect the system faults with the variations and noises in the input data. The use of CNN-LSTM eliminates the complicated and often manual feature extraction procedure that is commonly required by conventional data-driven approaches. In order to address the issue of insufficient training data in practice, a transfer learning method is also applied to facilitate the future practical applications. The efficacy of the proposed data-driven approach is tested for the protection gaps with respect to the transmission line high-impedance fault and transformer inter-turn fault, respectively. The extensive study results demonstrate that the proposed approach can effectively bridge the protection gaps in power system operations.

1. Introduction

Electric power grid is subject to faults and failures that can result in unsafe conditions for humans and damage power equipment or even cause the entire system to break down. The goal of power system protection is to isolate the faults or the affected components as soon as possible without interrupting the capability of the system in serving electric loads [1]. Therefore, the protection schemes must accurately distinguish system faults from system transients as a result of normal operations such as capacitor bank switching, load changes, motors starting, and etc.

The key component of the traditional power system protection technology is the protective relay that monitors the system conditions and makes decisions to interrupt the circuits if faults are detected. Existing protective relays are often designed to issue tripping commands when the preset thresholds are exceeded [2]. However, it is sometimes very difficult to determine the accurate thresholds because their selections depend on many factors such as operating conditions, equipment parameters, system transients, and fault types. Therefore, the threshold settings in practice usually represent a trade-off between the protection sensitivity and security, and protective relays cannot provide reliable or secure protection against faults or transients under all circumstances

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[3]. “Protection gap” is a term that describes the inadequacies in protection schemes where the existing technology would either be incapable of detecting the existence of a fault, or would misjudge normal operations and inadvertently perform a tripping action when such action is not desired. For example, when some faults happen, the monitored system measurements may resemble normal conditions. Then, the relays are not sensitive enough to detect this type of fault, which could result in damages to power components and possibly even system blackouts.

Among various protection gaps, the ones with respect to the transmission line high-impedance fault (HIF) [4] and transformer inter-turn fault [5] are very common but dangerous. The transmission line HIF usually occurs when a quasi-conductive object (such as a tree branch) contacts the line, or the transmission line breaks and falls to the ground. Because the quasi-conductive objects or ground have high resistance, the induced fault current is usually very low. Therefore, it is very difficult for traditional relaying technologies to detect the existence of a transmission line HIF. However, the transmission line HIF represents a great threat to the personnel and public safety. The voltage around the downed conductor is very high, and thus, people who walk near the downed conductor could be shocked if they were unaware of the hazard. In addition, the constant arcing caused by the transmission line HIF would easily ignite wild fires in the forests during dry and hot seasons. In practice, it accounts for around 5–20% of distribution faults, so they represent a serious public safety hazard [6].

In order to detect transmission line HIFs, it have been proposed to measure both the voltages and currents on the transmission line and calculating the third-harmonic current phase angle with respect to the fundamental voltage [6]. When a transmission line HIF occurs, the measured values is vectorially subtracted from the previous stored normal values. The relay issues a tripping command if the magnitude of the value is above a certain threshold. However, with the increasing volume of inverter-based distributed energy resources (DERs) and induction motors, various third-harmonic currents exist at all levels, which represents a serious security issue for the harmonic-based protection method [7]. Another approach to transmission line HIF detection applies an artificial neural network (ANN) along with traditional data feature extraction methods like frequency analysis or wavelet transform (see, for example, [6], [8] and [9]). This approach aims to capture both the frequency and instant (location) information by mapping the time-series voltage and/or current waveform into the time domain. Then an ANN is trained to find the relationship between the frequency and instant information and the corresponding system conditions. The shortcoming of this approach is that the feature extraction requires significant human efforts, and the parameters and settings of the approach would differ if the system structure or operation states were changed.

Transformer inter-turn fault is another fault that has led to the protection gap for decades. During the normal operation of power transformers, the windings and iron cores generate heat because of losses in conduction, hysteresis, and eddy currents. This heat would slowly damage the insulation of the windings over years of operation, unnoticed by the protection relays. If the winding insulation is worn out, one or more sequential pairs of turns will be shorted, which causes the “inter-turn” (or turn-turn) faults [10]. However, the low fault currents make these inter-turn faults extremely difficult for the protective relaying technology to detect them. When an inter-turn fault starts to occur inside the transformer, it is often the case that only a very small portion of the windings is shorted. Therefore, the primary-secondary turn ratio and the magnetic flux will remain almost unchanged, resulting in negligible changes in terminal voltage and current measurements [11]. However, because the short resistance is so low, the fault current through the shorted circuit could reach thousands of amperes. The extremely high fault currents generate copious heat and cause local thermal overloading, which ultimately evolves to catastrophic failure if the inter-turn fault is not detected and isolated in its earliest stage [12].

The most popular approach to inter-turn fault detection is the

negative-sequence differential protection scheme [13]. Inter-turn faults give rise to asymmetries in transformer terminal currents, so the negative-sequence currents at transformer terminals are good indicators of inter-turn faults. However, studies have shown that this scheme is not sensitive enough to detect some inter-turn faults involving only very few turns [14]. Recent studies in [15] and [16] have shown that the combination of ANN and feature extraction methods can help detect inter-turn faults. The concepts behind these approaches are similar to those used for transmission line HIF detection as discussed previously, and thus they share similar shortcomings.

In this paper, a new data-driven approach is proposed to bridge power system protection gaps. Instead of replacing the existing relaying technologies, the proposed data-driven approach will complement operative relays by specifically targeting the protection gaps. The goal is to increase the overall system operational resilience by introducing additional data-driven technology to work alongside the traditional protection technology. The proposed data-driven approach builds and trains a transfer-learning oriented deep neural network that utilizes convolutional neural network (CNN) [17] and long short-term memory (LSTM) [18] to find the mapping from raw instrumental measurements to the corresponding faulty or normal operating conditions. The effectiveness of this approach is illustrated by successfully addressing the protection gaps with respect to the transmission line HIF and transformer inter-turn fault.

As any existing data-driven ones, the proposed approaches can capture differences in the magnitude and pattern of the measurements under normal operations and fault conditions when compared to traditional relays that are based on preset thresholds. Furthermore, it overcomes three major limitations of conventional data-driven approaches. First, a conventional ANN does not take into account the spatial structure of data, and it is not very effective in obeying translation invariance. In contrast, the convolutional and pooling layers of a CNN help preserve the translation invariance, while the LSTM is very effective for characterizing the temporal correlation of time-series data. Hence, by using the CNN-LSTM, the proposed approach can accurately detect a fault even if it varies. Second, many conventional ANN-based approaches require a complicated, time-consuming signal feature extraction procedure that uses either a discrete Fourier transform or a discrete wavelet transform to pre-process the raw input data. However, the CNN-LSTM can learn the signal features automatically during the training process so it eliminates the need to manually perform the complicated feature extraction. Third, the training of an ANN usually requires a large amount of data. Hence, it is very difficult to apply ANN-based approaches in practice with little training data. The proposed data-driven approach applies a transfer learning method to address this issue by leveraging a previously trained CNN-LSTM that has captured certain fault characteristics and patterns from simulation data.

The remaining of this paper is organized as follows. In Section 2, the modeling of both transmission line HIF and transformer inter-turn fault is first briefly introduced. Then, a new data-driven approach to bridging protection gaps is proposed in Section 3, and then tested and verified against the transmission line HIF and transformer inter-turn fault in Section 4. Conclusions can be found in Section 5.

2. Fault modeling for protection gaps

In this paper, the protection gaps with respect to the transmission line HIF and transformer inter-turn fault are considered to evaluate and validate the performance of the newly proposed data-driven approach. In the following subsections, the modeling of these two faults is first briefly introduced.

2.1. Transmission line HIF

A generic model of the transmission line HIF [6,19–21] that contains antiparallel DC sources, diodes, and variable resistors is shown in Fig. 1,

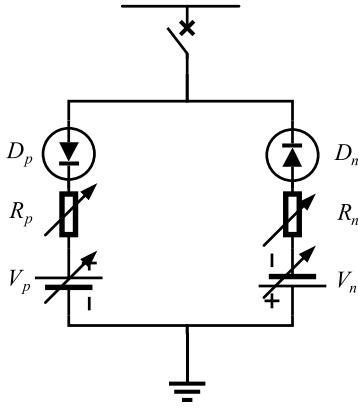


Fig. 1. Illustration of a generic model of transmission line HIF [6,19–21].

where the subscripts p and n stand for *positive* and *negative*, respectively. In order to simulate the transmission line HIF, the model in Fig. 1 is connected to the instantaneous phase voltage $V_{ph}(t)$. When $V_{ph}(t)$ is larger than the positive DC voltage V_p , the positive cycle of fault current flows toward ground through the left-hand path. When $V_{ph}(t)$ is smaller than the negative DC voltage V_n , the negative cycle of fault current flows from the ground through the right-hand path. Otherwise, neither the left nor the right path conducts. In simulation, the values of V_p , V_n , R_p , and R_n are randomly set to reflect the realistic characteristics of transmission line HIFs such as the nonlinear impedance, time-varying parameters, and random nature.

In Fig. 2, it shows an example of the simulation results using the generic model of the transmission line HIF, where the fault occurs at 18 ms. It can be seen from Fig. 2(a) that the nonlinear fault current only conducts during the peak or valley of the phase voltage (i.e., when $V_{ph}(t) > V_p$ or $V_{ph}(t) < V_n$). Therefore, the magnitude of the line current measured by the transmission line terminal relays changes little as shown in Fig. 2(b). In this case, a traditional overcurrent relay is not capable of detecting the simulated transmission line HIF. However, it

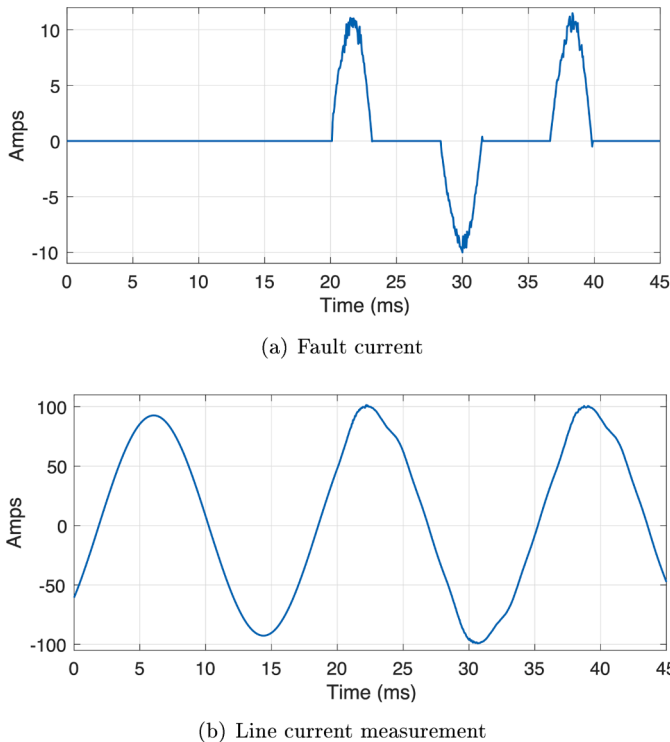


Fig. 2. Results from a simulated transmission line HIF.

can be seen from Fig. 2(b) that the transmission line HIF causes minor distortions to the sinusoidal current waveform. It is promising to apply the data-driven method that can capture the fault pattern and differentiate the fault from the normal transients.

2.2. Transformer inter-turn fault

The model of the transformer inter-turn fault is based on a single-phase, saturable-core transformer model [14,22] as shown in Fig. 3, where two additional branches are added to the transformer windings on the primary or secondary side. The fault location is determined by the parameters α and β , and the fault severity is determined by the value of $|\alpha - \beta|$ and the fault impedance in the short circuit.

It is easy to construct a delta-wye or wye-delta three-phase transformer by connecting three single-phase transformers shown in Fig. 3. In Fig. 4, it shows an example of the terminal measurements of a 40 MVA 115/35 kV three-phase transformer with a 1% inter-turn fault on the secondary side. It can be seen that it is hard to notice any change in the voltage or current measurements when the inter-turn fault occurs because the overall primary-secondary ratio stays almost the same. A traditional protective relay using percentage differential protection or negative-sequence differential protection methods [14] cannot detect such a small inter-turn fault.

3. New data-driven approach

In order to complement the traditional relays for improved power system protection, a new data-driven approach is proposed in this section to address the issues associated with those conventional ones. The proposed approach builds and trains a CNN-LSTM that can accurately distinguish faulty conditions from those normal transients, and then applies the transfer learning method to facilitate the transition of the trained CNN-LSTM based on fault simulations to practical applications.

3.1. CNN-LSTM

In order to overcome the identified shortcomings of traditional ANN-based methods, a deep neural network that combines the CNN and LSTM as shown in Fig. 5 is proposed for fault detection. The input data are raw terminal time-series measurements of those protection zones, which are either the transmission lines when the HIF is concerned or the transformers when the inter-turn fault is concerned.

The first three layers of the proposed deep neural network are CNN layers. Each of the CNN layers consists of a convolution operation, a rectified linear unit (ReLU) [23], and a max-pooling function. The convolution operation aims to capture the shape of the time-series data, and is mathematically defined as

$$(f * g)[n] = \sum_{i=1}^n f[i] \times g[i], \quad (1)$$

where f is the filter feature, g is the input that corresponds to the filter, and n is the size of the filter. The output of the convolution operation is then passed into a ReLU activation. The ReLU handles the vanishing-gradient problem quite well, and it is less computationally expensive than traditional hyperbolic tangent and sigmoid activation operations. The output of the ReLU is defined as

$$z = \max\{0, (f * g)[n]\}. \quad (2)$$

A max-pooling operation follows the ReLU to reduce the dimensionality of the data and the neural network. It selects the maximum value of the input z in each pooling region and passes only the maximum values to the next layer. The max-pooling function allows filters in deeper layers to learn a general overview of the input patterns, as shown in Fig. 5. Therefore, the three CNN layers capture the data patterns at different levels, which means the neural network learns the details and overview

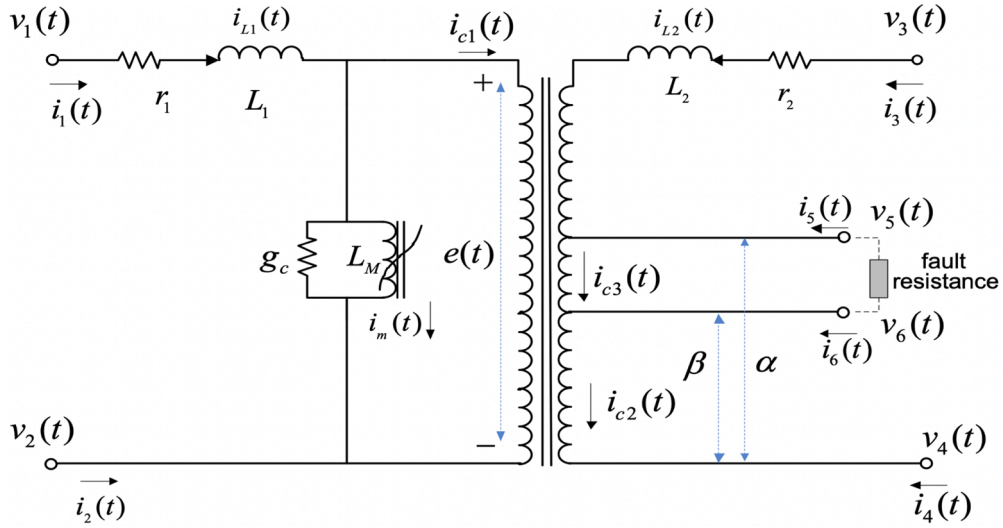


Fig. 3. Illustration of the model of transformer inter-turn fault [14,22].

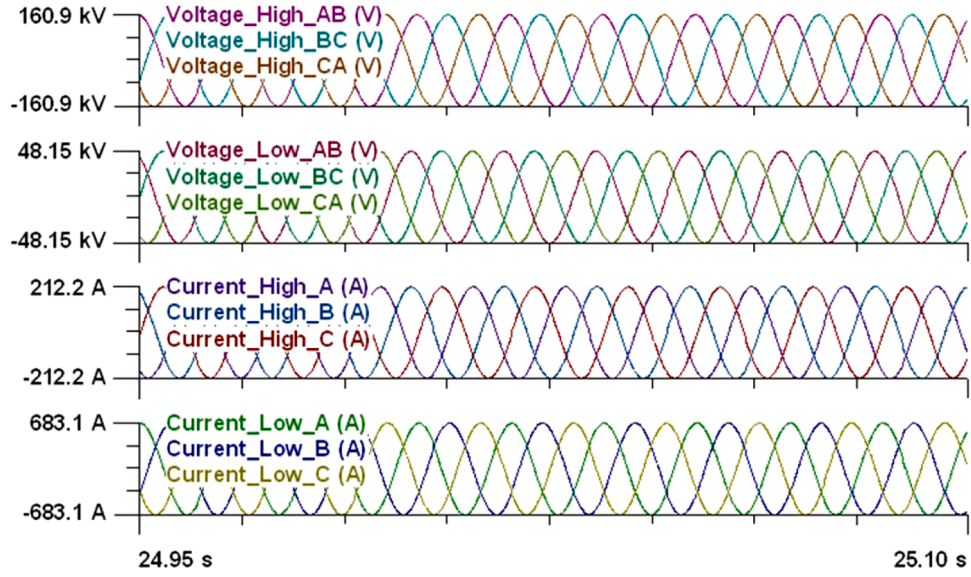


Fig. 4. Terminal measurements from a simulated transformer inter-turn fault.

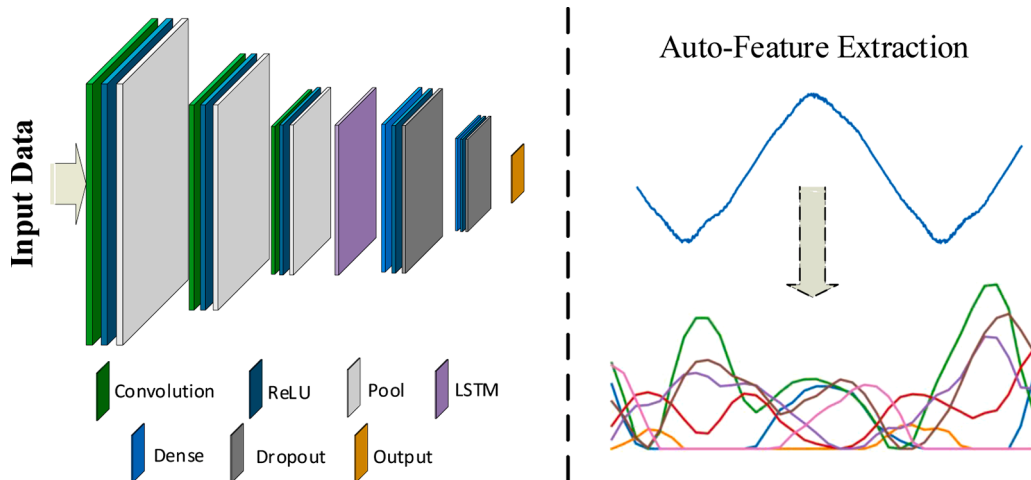


Fig. 5. Illustration of proposed deep neural network with CNN and LSTM, and auto-feature extraction.

at the same time.

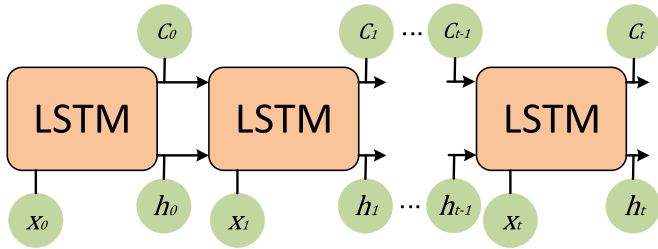
Following three CNN layers, an LSTM layer is connected. The LSTM is an advanced deep learning technique with the recurrent neural network (RNN) architecture, which performs well on processing time-series data by learning long-term dependencies [24]. The structure of the LSTM layer is shown in Fig. 6(a). Like the traditional RNNs, the LSTM has the form of a chain of repeating LSTM cells, and allows the outputs of the preceding cell to be used as the inputs of the subsequent while maintaining the hidden states h_t . In this way, the LSTM remembers the past information and its predictions now are influenced by both knowledge learned from the past and the current input. As a result, the LSTM is mostly applied to time series and sequences. Unlike the traditional RNNs, where each repeating cell has a simple structure, the LSTM cell has several functions that interact in a different manner as shown in Fig. 6(b). The cell state C_t that runs through the entire chain from one cell to the next can be modified by three functions that are referred to as gates using the previous states (C_{t-1} and h_{t-1}) as well as the current inputs (X_t). For the first cell, the cell and hidden states (C_0 and h_0) are usually by default. Three gates include the forget gate, input gate, and output gate, respectively. The forget gate is a sigmoid function with the previous hidden state h_{t-1} and current input X_t as the input. Mathematically, it is defined as

$$f_t = \sigma(W_f^h \times h_{t-1} + W_f^x \times X_t + b_f), \quad (3)$$

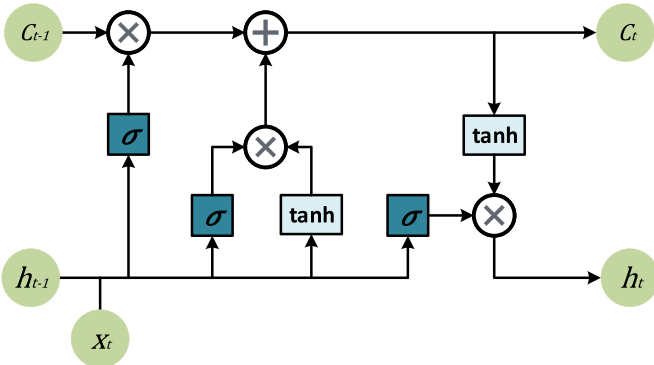
where $\sigma(\cdot)$ denotes the sigmoid function, W_f^h , W_f^x and b_f are the gate parameters. The output of the forget gate is either 1 or 0, which implies that the previous cell state C_{t-1} is to be remembered or forgotten, respectively. Following the forget gate, the input gate determines the new information to be included into the cell state by multiplying two quantities together. The first quantity is defined as

$$i_t = \sigma(W_i^h \times h_{t-1} + W_i^x \times X_t + b_i), \quad (4)$$

and the second one as



(a) LSTM chain



(b) Single LSTM cell

Fig. 6. Illustration of the LSTM layer in the proposed deep neural network.

$$\hat{C}_t = \tanh(W_c^h \times h_{t-1} + W_c^x \times X_t + b_c), \quad (5)$$

where W_i^h , W_i^x , b_i , W_c^h , W_c^x and b_c are the parameters of the input gate. Then, the cell state is updated as

$$C_t = f_t \times C_{t-1} + i_t \times \hat{C}_t. \quad (6)$$

Finally, the output gate updates the hidden state for the next LSTM cell as

$$h_t = o_t \times \tanh(C_t) \quad (7)$$

with

$$o_t = \sigma(W_o^h \times h_{t-1} + W_o^x \times X_t + b_o), \quad (8)$$

where W_o^h , W_o^x and b_o are the parameters of the output gate. These special gates of each LSTM cell can help the network to forget the old information or update the new information, which help bridge long temporal lags in the input data.

Dense layers that are connected to the end of the CNN-LSTM network with dropout regularization [25] are used to prevent overfitting problems. The final output layer uses a sigmoid activation function to determine whether the input data correspond to the faulty or normal transients.

After the outputs of the proposed deep neural network for all the given input data are obtained, the following cross-entropy is defined as the loss function,

$$J = -\frac{1}{M} \sum_{j=1}^M (y_j^* \log(y_j) + (1 - y_j^*) \log(1 - y_j)), \quad (9)$$

where y_j^* and y_j are the true and assigned labels of the j -th input data, respectively, and M is the number of input data. The proposed deep neural network updates the parameters (or weights) of each layer with the objective of minimizing the loss J through a back-propagation process [26].

Remark 1. The CNN and LSTM are in a class of models that are both spatially and temporally deep. In this paper, the CNN is used as a data “encoder” that automatically captures the patterns of the input data at different levels, and the LSTM as a “decoder” that generates the output to differentiate different patterns with the corresponding labels (faulty or normal). Unlike the traditional ANN-based method, the combined CNN-LSTM network achieves the data translation invariance and captures the temporal correlation of the time-series input data. In addition, the use of CNN and LSTM eliminates the complicated and manual feature extraction that is time consuming but often required by many conventional data-driven approaches, as shown in Fig. 5.

3.2. Transfer learning

It usually requires a large amount of data to train the neural network for ANN-based approaches. However, the availability of training data in practice is often limited, which presents a significant barrier for the practical applications of ANN-based approaches. In order to address this problem, the transfer learning technique [27] is adopted by the proposed data-driven approach. The basic concept of transfer learning is illustrated in Fig. 7. Suppose there are two different learning tasks (I and II). Instead of training a neural network from the scratch with random initialization, the learning experience from Task I can be leveraged and transferred to the training process in Task II.

Remark 2. Transfer learning is just like telling ducks eggs from ducks without much specific information after learning the difference between chicken eggs and chickens. Although Task I (chicken eggs versus chickens) and Task II (duck eggs versus ducks) are different, these two

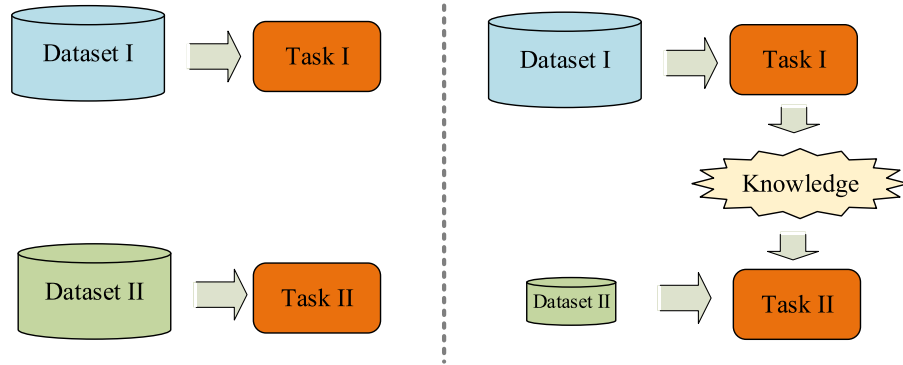


Fig. 7. Traditional learning versus transfer learning.

tasks share some common features and patterns. Hence, the knowledge learned from Task I can be easily used to complete Task II.

The solution architecture diagram of the proposed data-drive approach is shown in Fig. 8. In order to facilitate the practical applications, the proposed CNN-LSTM is first trained over a simulated system with abundant training data so that it can capture the features and patterns of protection gaps. Then, the trained CNN-LSTM can be used for online protection of a zone (transformer, lines, etc.) with partially frozen parameters to address similar or identical practical protection gaps via transfer learning if the real world data is limited. When moving from the simulated system to the real system, the trained neural network will be fine-tuned with minimal changes. Therefore, the corresponding training process requires fewer training data with faster training speed than those without the use of transfer learning. The CNN-LSTM will continuously monitoring the two-cycle input data through a sliding window (user can decide the stride), and it will make a trip decision if it detects the existence of a fault.

3.3. Performance metrics

The proposed data-driven approach to bridging protection gaps is actually a binary classification method, whose output (i.e., the label of the input data) is either 1 (*Positive*) or 0 (*Negative*) denoting the faulty and normal conditions, respectively. The most popular metric for evaluating the performance of such an approach is ACCURACY that is

defined as the ratio of the number of true predictions to the total number. However, this performance metric does not provide enough information about false predictions, which could be a problem when there are much more (or less) positive labels than negative labels. In order to reasonably evaluate the performance of data-driven approaches, F1 SCORE is another performance metric to be used herein.

As pointed out in the above, the label of the binary classification is either *Positive* or *Negative*, and the corresponding prediction can be either *True* or *False*. Therefore, the prediction results of the binary classification can be classified into four different categories including (1) *True Positive* (T_P) when the assigned label y is *Positive* and the true label y^* is also *Positive*; (2) *True Negative* (T_N) when y is *Negative* and y^* is also *Negative*; (3) *False Positive* (F_P) when y is *Positive* but y^* is *Negative*; (4) *False Negative* (F_N) when y is *Negative* but y^* is *Positive*. It follows that the performance metric ACCURACY can be defined as

$$\text{ACCURACY} = \frac{T_P + T_N}{T_P + T_N + F_P + F_N}. \quad (10)$$

In order to similarly define the performance metric F1 SCORE, it is necessary to define two additional quantities referred to as PRECISION and RECALL first. PRECISION is the ratio of the *True Positive* prediction results to the total *Positive* prediction results, which can be mathematically defined as

$$\text{PRECISION} = \frac{T_P}{T_P + F_P}. \quad (11)$$

RECALL is the ratio of the *True Positive* prediction results to the number of input data whose true labels are *Positive*, which can be mathematically defined as

$$\text{RECALL} = \frac{T_P}{T_P + F_N}. \quad (12)$$

Then it follows that the performance metric F1 SCORE can be defined as

$$\text{F1 SCORE} = \frac{2 * \text{PRECISION} * \text{RECALL}}{\text{PRECISION} + \text{RECALL}}. \quad (13)$$

It can be seen from the above that ACCURACY is only related to T_P and T_N whereas F1 SCORE also takes into account F_P and F_N for a more comprehensive performance evaluation. In the following case studies, both ACCURACY and F1 SCORE are applied as the performance evaluation metrics.

4. Case studies

In this paper, the performance of the proposed data-driven approach

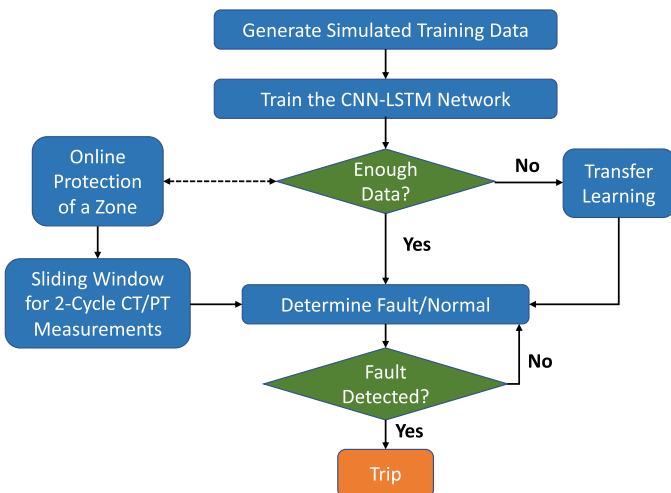


Fig. 8. Solution architecture diagram of proposed approach.

in detecting transmission line HIFs and transformer inter-turn faults is compared with that of traditional data-driven approaches. In particular, the support vector machine (SVM) in [8] and the multilayer perceptron (MLP) neural network in [9,16] are tested and compared to the proposed CNN-LSTM. Unlike the proposed CNN-LSTM, which can take the raw data directly, the application of the SVM and MLP neural network require the preprocessing of the input data.

For the SVM, a wavelet transform is first used to capture the time and frequency information, and then a radial basis function kernel to build the classifier. For the MLP neural network, a frequency analysis is first used to get the harmonics and magnitudes, and then a four-layer neural network with 256, 256, 128, and 64 neurons in the respective layers to discriminate fault and normal transients. For the proposed CNN-LSTM, the first three CNN layers use 32, 64, and 128 1-D filters with a size of seven, and both the LSTM and dense layer have 128 units.

4.1. Case one: HIF

In this case study, the transmission line HIF model in Section 2 is implemented into the IEEE 34-bus, 24.9 kV test feeder [28]. We used the ATP-EMTP software to build the simulation system, and used Python to automatically change the system parameters and run thousands of simulation cases. Around 4,500 operating scenarios are simulated to generate both faulty transients from transmission line HIFs and normal transients from various events such as load changes, sectionalizer switching, capacitor bank changes, DER generation, induction motor operation, and transformer tap changes. For the transient simulation, various system conditions are considered to reflect different system loading levels (10%–90%), voltage levels (0.95 p.u.–1.05 p.u.), event locations, transient inception angles, and line phases. The resistance of the HIF resistor varies between 150 and 700 ohms, and the conducting voltage between 65% and 85%. The input data for data-driven approaches are selected to be the time-domain sampled current measurements from the feeder terminal with the sampling rate of 15 kHz and the data length of two cycles. In addition, 2% white noise is added to the input data to mimic the realistic measurement noise.

From those generated transients, around 3,600 scenarios are used for training and the remaining 900 scenarios are retained for testing. For the training of the MLP neural network and CNN-LSTM, 10% of the training data are used for validation in order to prevent overfitting. The cross-entropy losses of the MLP neural network and CNN-LSTM during the training process are shown in Fig. 9, respectively. Note that the SVM does not use the training and validation algorithms as neural networks. It can be seen from Fig. 9 that the cross-entropy losses of both neural networks decrease as the training progresses and finally reach the steady states. The cross-entropy losses from the MLP neural network are greater than those from the proposed CNN-LSTM, indicating that the CNN-LSTM outperforms the MLP neural network in differentiating transmission line HIFs from normal transients. The testing results as shown in Table 1 also

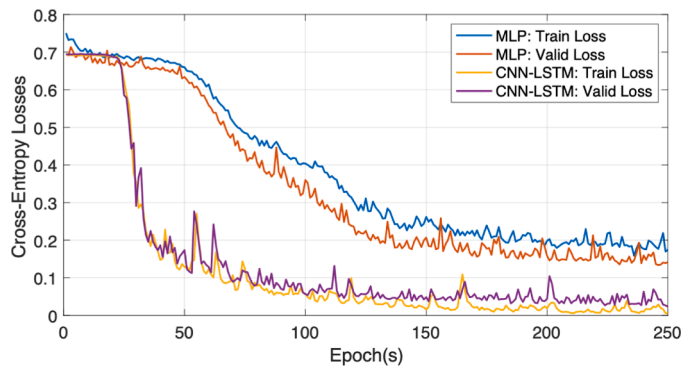


Fig. 9. Cross-entropy loss comparison between the MLP neural network and CNN-LSTM in Case One.

Table 1

Testing result comparison for detecting transmission line HIFs in Case One.

	SVM	MLP	CNN-LSTM
True Positive T_P	366	370	393
False Positive F_P	16	17	3
False Negative F_N	34	30	7
True Negative T_N	487	486	500
Accuracy	94.5%	94.8%	98.9%
F1 score	93.6%	94.0%	98.7%

confirm the effectiveness of the proposed CNN-LSTM. The ACCURACY of the SVM and MLP neural network is 94.5% and 94.8%, and the F1 SCORE is 93.6% and 94.0%, respectively. In contrast, the CNN-LSTM has a much higher ACCURACY of 98.9% and F1 SCORE of 98.7%. Therefore, the proposed data-driven approach can better detect transmission line HIFs than traditional approaches.

4.2. Case two: Inter-turn fault

In this case study, the transformer inter-turn fault model in Section 2 is implemented with a 20 MVA 115/25 kV three-phase saturable-core transformer. We used the WinIGS software to build the simulation system, and used python to automatically change the system parameters and run thousands of simulation cases. Around 3,000 operating scenarios are simulated to generate both faulty transients from transformer inter-turn faults and normal transients from various events including load changes, capacitor bank changes, DER generation, and induction motor operation. The transformer inter-turn fault location varies on both the primary and secondary windings with a fault level between 1% and 8%. The fault current in the shorted winding are extremely high from 700 to thousands of Amperes, while the current change at the terminal are very low from 4 to 40 amperes. Besides those within the transformer protection zone, external adjacent line faults with randomly selected fault type, fault phase, and fault location, outside the protection zone are also simulated. Because the transients corresponding to external faults do not trip the transformer breaker, they are also included into the normal transients. The input data for data-driven approaches are selected to be the time-domain sampled phase voltage and current measurements of the transformer. As in Case One, the same sampling rate and data length are also used, and the same amount of white noise is added to the input data.

From those generated transients, around 2,400 scenarios are used for training and the remaining 600 scenarios are retained for testing. The comparison results as shown in Fig. 10 and Table 2 show that proposed CNN-LSTM still outperforms the SVM and MLP neural network in differentiating transformer inter-turn faults from normal transients. The ACCURACY of the SVM and MLP neural network is 94.8%, and 95.6%, and the F1 SCORE is 93.9% and 94.7%, respectively, while the CNN-LSTM has the ACCURACY of 99.3% and F1 SCORE of 99.2%.

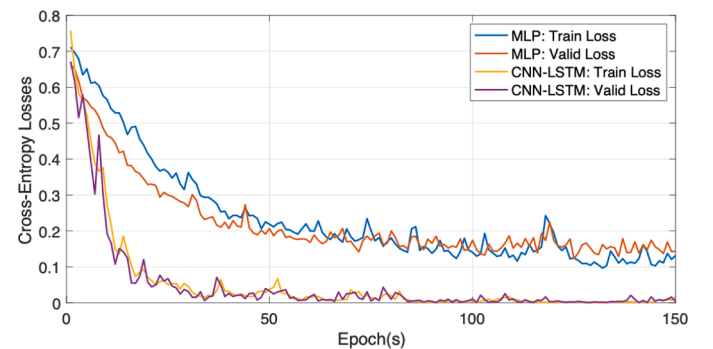


Fig. 10. Cross-entropy loss comparison between the MLP neural network and CNN-LSTM in Case Two.

Table 2

Testing result comparison for detecting transformer inter-turn HIFs in Case Two.

	SVM	MLP	CNN-LSTM
True Positive T_P	248	239	255
False Positive F_P	25	11	4
False Negative F_N	7	16	0
True Negative T_N	330	344	351
Accuracy	94.8%	95.6%	99.3%
F1 score	93.9%	94.7%	99.2%

Table 3

Test results for transfer learning and for random initialization

	Random initialization	Transfer learning
True Positive T_P	34	63
False Positive F_P	23	1
False Negative F_N	31	2
True Negative T_N	31	53
Accuracy	54.6%	97.5%
F1 score	55.7%	97.7%

4.3. Case three: Transfer learning

In the above case studies, a large amount of data has been made available for the training of the proposed CNN-LSTM for effective fault detection. However, there may not be enough training data in practice, which necessitates the transfer learning method introduced in Section 3. In this case study, transfer learning is illustrated to facilitate the practical applications of the proposed CNN-LSTM when the amount of training data is greatly limited.

Specifically, the proposed CNN-LSTM is applied to detect transmission line HIFs in the IEEE 13-bus, 4.2 kV test feeder [28], where a different transmission line HIF model from the one in Section 2 is simulated. Instead of the antiparallel sources, diodes, and variable resistors as shown in Fig. 1, the transient analysis of control systems in [29] is used for the transmission line HIF models. The fault impedance varies from 10 to 200 ohms, and the conducting voltage from 60% to 80%.

In the following study, this IEEE 13-bus test feeder serves as the “real” system while the IEEE 34-bus test feeder in Case One as the simulated system. Given the different test system and fault model, the corresponding input data from both systems are different. For the “real” system, only around 240 operating scenarios are generated to limit the availability of training data. About half of the data are used to training, of which 10% is again reserved for validation, and the remaining for testing. The performance of the proposed CNN-LSTM in detecting transmission line HIFs is then examined with and without transfer learning, respectively. Without transfer learning, the CNN-LSTM will be trained from the scratch via random initialization based on the data from the “real” system only. With transfer learning, the weights and network structure are copied from the Case One, which means the knowledge and features learned from Case One is leveraged. In other words, the trained CNN-LSTM from Case One is fine-tuned using the new training data from the “real” system. The cross-entropy losses of the CNN-LSTM with and without transfer learning are shown in Fig. 11, respectively. The training is terminated right before overfitting occurs for the best performance of both methods. It is clearly shown that the use of transfer learning results in much smaller cross-entropy losses. The testing results shown in Table 1 also confirm the effectiveness of transfer learning. The ACCURACY and F1 SCORE without transfer learning are as low as 54.6% and 55.7%, respectively. It implies that the proposed CNN-LSTM could not capture the features of transmission line HIFs with very limited data, which actually leads to random guessing. On the contrary, the CNN-LSTM with transfer learning ends up with a much higher ACCURACY of 97.5% and F1 SCORE of 97.7%. Therefore, transfer learning has the great potential to overcome the challenge of limited training data

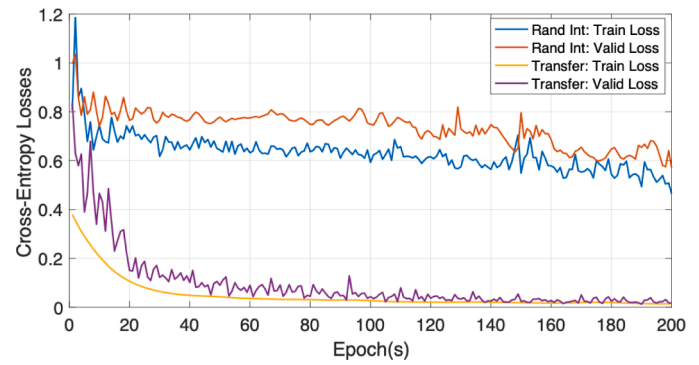


Fig. 11. Cross-entropy losses of the CNN-LSTM with and without transfer learning in Case Three.

in practice.

5. Conclusions

In this paper, a new data-driven approach based on deep learning was proposed to bridge existing power system protection gaps. The proposed approach applies a CNN-LSTM for accurate fault detection. Detailed case studies illustrated the better performance of the proposed approach over traditional data-driven approaches in detecting transmission line HIFs and transformer inter-turn faults. The proposed approach also incorporates transfer learning to effectively address the challenge of insufficient training data in practice, and greatly facilitate the future practical applications.

CRedit authorship contribution statement

Rui Fan: Conceptualization, Methodology, Software, Writing – original draft. **Tianzhixi Yin:** Methodology, Formal analysis, Investigation, Visualization. **Kun Yang:** Validation, Writing – review & editing. **Jianming Lian:** Supervision, Writing – review & editing, Project administration. **John Buckheit:** Software, Data curation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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